

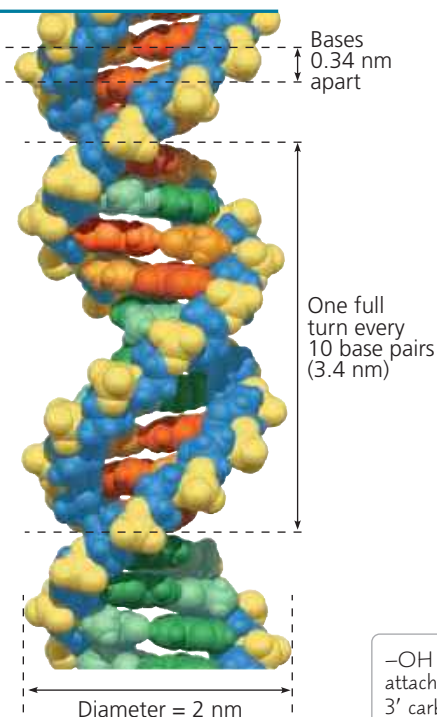
▼ Figure 16.7 VISUALIZING DNA

DNA can be illustrated in many ways, but all diagrams represent the same basic structure. The level of detail shown depends on the process or the type of information being conveyed.

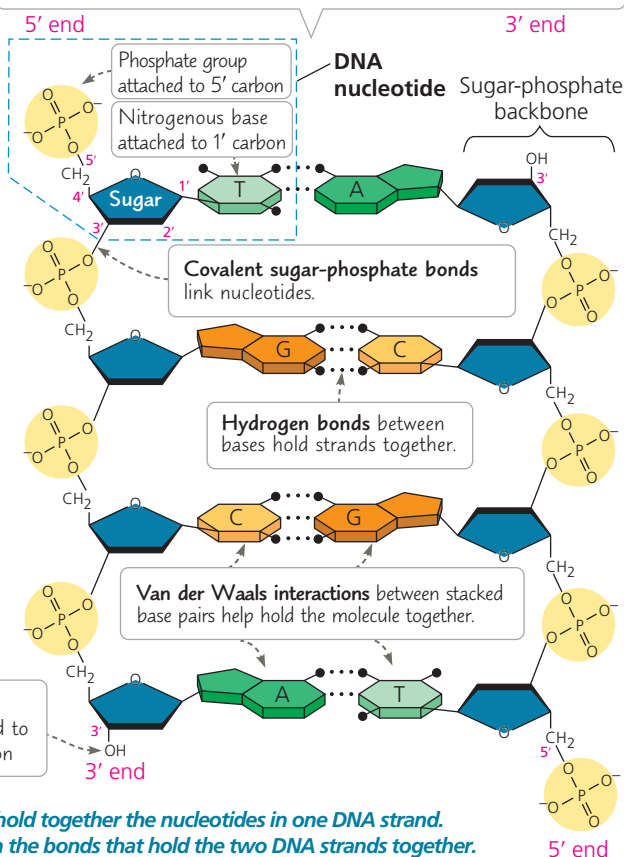
Structural Images

The space-filling model on the left shows the 3-D shape of the DNA double helix; the diagram on the right shows chemical details of DNA's structure. In both images, phosphate groups are yellow, deoxyribose sugars are blue, and nitrogenous bases are shades of green and orange.

The DNA double helix is "right-handed." Use your right hand as shown to follow the sugar-phosphate backbone up the helix and around to the back. (It won't work with your left hand.)



Here, the two DNA strands are shown untwisted so it's easier to see the chemical details. Note that the strands are **antiparallel**—they are oriented in opposite directions, like the lanes of a divided street.

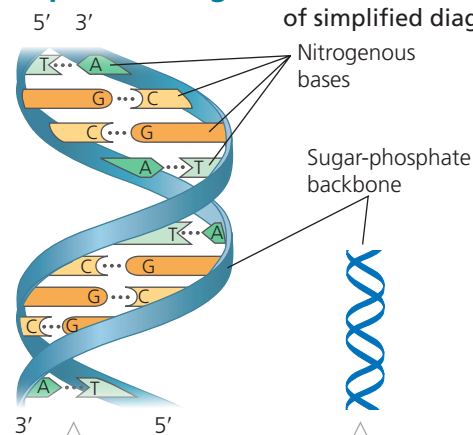


➔ Mastering Biology Animation: DNA Double Helix

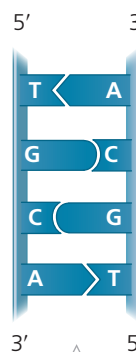
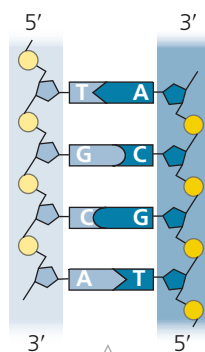
1. Describe the bonds that hold together the nucleotides in one DNA strand. Then compare them with the bonds that hold the two DNA strands together.
2. How do the two ends of a DNA strand differ in structure?

Simplified Images

When molecular detail is not necessary, DNA is portrayed in a range of simplified diagrams, depending on the focus of the figure.



The "ribbons" in these simplified double helix diagrams represent the sugar-phosphate backbones; these models emphasize the 3-D nature of DNA.



Sometimes the double-stranded DNA molecule is shown simply as two straight lines.

3. Compare the information conveyed in the three ladder diagrams.

DNA Sequences

Genetic information is carried in DNA as a linear sequence of nucleotides that may be transcribed into mRNA and translated into a polypeptide. When focusing on the DNA sequence, each nucleotide can be represented simply as the letter of its base: A, T, C, or G.

3' - A C G T A A G C G G T T A A T - 5'
5' - T G C A T T C G C C A A T T A - 3'

➔ **Instructors:** Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.